Lecture 12 (*March 15*)

Reading: Section 4.2
Viewing: Lecture 15

We will now give a rigorous definition of projections.

Definition 13. *Given a vector $\boldsymbol{b} \in \mathbb{R}^n$ and a linear subspace $V \subset \mathbb{R}^n$, the* ***(orthogonal) projection*** *of $\boldsymbol{b}$ onto V is the unique vector:*

$$\boldsymbol{p} = \text{proj}_V \boldsymbol{b} \in V$$

such that $(\boldsymbol{b} - \boldsymbol{p}) \perp V$. Put differently, $\boldsymbol{p}$ is the closest vector in V to $\boldsymbol{b}$.

Intuitively, suppose you want to measure a quantity, which general theory tells you must be a vector $\boldsymbol{p}$ in a subspace $V \subset \mathbb{R}^n$. However, your measurement will probably have an error, and you might get a value $\boldsymbol{b}$ outside of V. Then your best guess for the actual value $\boldsymbol{p} \in V$ is to set it equal to the orthogonal projection of $\boldsymbol{b}$ on V, since this is the closest vector in V to the measured value $\boldsymbol{b}$. That is also why the vector $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{p}$ is sometimes called the **error**.

Let us work out formulas for the projection of a vector $\boldsymbol{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$. The easiest case is when V is one dimensional, i.e. the line spanned by a vector $\boldsymbol{a} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$. Then the orthogonal projection $\boldsymbol{p}$ must be a multiple of this vector, so we are looking for:

$$\boldsymbol{p} = \lambda \boldsymbol{a} \quad \text{such that} \quad (\boldsymbol{b} - \lambda \boldsymbol{a}) \perp \boldsymbol{a}$$

Using the dot product interpretation of perpendicularity, we need:

$$0 = (\boldsymbol{b} - \lambda \boldsymbol{a}) \cdot \boldsymbol{a} \quad \Rightarrow \quad \lambda = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\boldsymbol{a} \cdot \boldsymbol{a}} = \frac{\boldsymbol{a}^T \boldsymbol{b}}{\boldsymbol{a}^T \boldsymbol{a}}$$

So the formula for the projection of $\boldsymbol{b}$ onto the line spanned by a single vector $\boldsymbol{a}$ is:

$$\text{proj}_{\boldsymbol{a}} \boldsymbol{b} = \boxed{\boldsymbol{a} \quad \underbrace{\frac{\boldsymbol{a}^T \boldsymbol{b}}{\boldsymbol{a}^T \boldsymbol{a}}}_{\text{this is a scalar}}} \tag{95}$$

Here's a sanity check: assume $\boldsymbol{a} = \boldsymbol{e}_i$ is the unit vector on the i–th axis. Then formula (95) reads:

$$\text{proj}_{\boldsymbol{e}_i} \boldsymbol{b} = \boldsymbol{e}_i b_i \tag{96}$$

so the projection only remembers the i–th component of the vector $\boldsymbol{b}$. It is easy to see that the operation (96) can be presented as a matrix multiplying the vector $\boldsymbol{b}$, specifically:

$$P_i \boldsymbol{b} = \boldsymbol{e}_i b_i \qquad \text{where} \qquad P_i = \begin{bmatrix} \dots & \dots & 0 & \dots & \dots \\ \dots & \dots & 0 & \dots & \dots \\ 0 & 0 & 1 & 0 & 0 \\ \dots & \dots & 0 & \dots & \dots \\ \dots & \dots & 0 & \dots & \dots \end{bmatrix} \tag{97}$$

with the unique 1 being at the intersection of row i and column i, and 0's everywhere else. The projection onto an arbitrary vector $\boldsymbol{a}$ can be described by a similar matrix, specifically:

$$P_{\boldsymbol{a}}\boldsymbol{b} = \boldsymbol{a}\frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}} \qquad \text{where} \qquad P_{\boldsymbol{a}} = \frac{\boldsymbol{a}\boldsymbol{a}^T}{\boldsymbol{a}^T\boldsymbol{a}} \tag{98}$$

Indeed, since $\boldsymbol{a}$ is an $n \times 1$ vector, the numerator of $P_{\boldsymbol{a}}$ is an $n \times n$ matrix, and the denominator is a 1×1 matrix, i.e. a scalar. This all fits into the following principle:

Fact 9. *For any subspace $V \subset \mathbb{R}^n$, there exists an $n \times n$ matrix P_V such that:*

$$\boxed{\text{proj}_V \boldsymbol{b} = P_V \boldsymbol{b}} \tag{99}$$

for any vector $\boldsymbol{b} \in \mathbb{R}^n$.

So all orthogonal projections are given by left multiplication with a suitable matrix (unoriginally called a "projection matrix"). The task for us now is to compute the matrix corresponding to a given projection, and we will do it in the special case when:

$$V = C(A)$$

for an $n \times m$ matrix A. In fact, this does not represent any restriction on the subspace V, since any subspace can be written as the column space of a suitably chosen matrix: just take basis vectors of the subspace and put them next to each other in a matrix. So given $\boldsymbol{b} \in \mathbb{R}^n$, we are looking for:

$$\boldsymbol{p} \in C(A) \quad \text{such that} \quad (\boldsymbol{b} - \boldsymbol{p}) \perp C(A)$$

But $\boldsymbol{p} \in C(A)$ precisely means:

$$\boldsymbol{p} = A\boldsymbol{v} \tag{100}$$

for some $m \times 1$ vector $\boldsymbol{v}$. Moreover, the condition $(\boldsymbol{b} - \boldsymbol{p}) \perp C(A)$ is equivalent to:

$$A^T(\boldsymbol{b} - \boldsymbol{p}) = 0 \tag{101}$$

because we know from the previous Lecture that a vector being orthogonal to the column space of A is equivalent to it lying in the left nullspace of A. Combining (100) with (101) gives us:

$$A^T(\boldsymbol{b} - A\boldsymbol{v}) = 0 \quad \Rightarrow \quad A^TA\boldsymbol{v} = A^T\boldsymbol{b}$$

Therefore, we may solve for $\boldsymbol{v} = (A^TA)^{-1}A^T\boldsymbol{b}$, and by combining this with (100) we obtain the following formula for the projection of the vector $\boldsymbol{b}$ onto the subspace $C(A)$:

$$\boxed{\text{proj}_{C(A)}\boldsymbol{b} = A(A^TA)^{-1}A^T\boldsymbol{b}} \tag{102}$$

Therefore, we conclude that the $n \times n$ projection matrix that Fact 9 predicted gives rise to the projection onto the subspace $C(A)$ is:

$$\boxed{P_{C(A)} = A(A^TA)^{-1}A^T} \tag{103}$$

If the case $m = 1$, i.e. A is a single $n \times 1$ vector, formula (103) is equivalent to (98). The $m \times m$ matrix $S = A^T A$ which appears in (103) is symmetric, but we need it to be invertible in order for the formulas presented above to hold. Luckily for us, we have the following fact:

$$\textbf{if the columns of } A \textbf{ are independent, then } S = A^T A \textbf{ is invertible} \tag{104}$$

When working with formulas such as (102) or (103), note that the equality:

$$(A^T A)^{-1} \neq A^{-1}(A^T)^{-1}$$

does not hold in general, because when A is a rectangular matrix, its inverse is not well-defined.

Let's do an example. Suppose we are in $n = 3$ dimensional space, and we want to compute the projection onto the plane V spanned by the vectors:

$$\begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \tag{105}$$

First of all, let's make sure that these vectors are linearly independent (if one were a linear combination of the other, we could just throw it out). Then, we put the vectors together in a matrix:

$$A = \begin{bmatrix} 2 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix}$$

As we saw in (103), an important role is played by the symmetric matrix:

$$S = A^T A = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}$$

whose inverse (this matrix is invertible because the vectors (105) are independent) is given by:

$$S^{-1} = \frac{1}{6} \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$$

Then formula (103) for the projection matrix onto the subspace spanned by the vectors (105) is:

$$P_V = \begin{bmatrix} 2 & -1 \\ 0 & 1 \\ -1 & 0 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5 & -1 & -2 \\ -1 & 5 & -2 \\ -2 & -2 & 2 \end{bmatrix} \tag{106}$$

We conclude that the projection of any vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ onto the given 2-dimensional plane is:

$$P_V \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5x - y - 2z \\ -x + 5y - 2z \\ -2x - 2y + 2z \end{bmatrix}$$

Remark. *Projection matrices are special in that they satisfy the equation:*

$$P_V P_V = P_V$$

for any subspace V. This corresponds to the geometric fact that the projection of a projection is the projection itself. Check that this equation holds for the specific example (106).